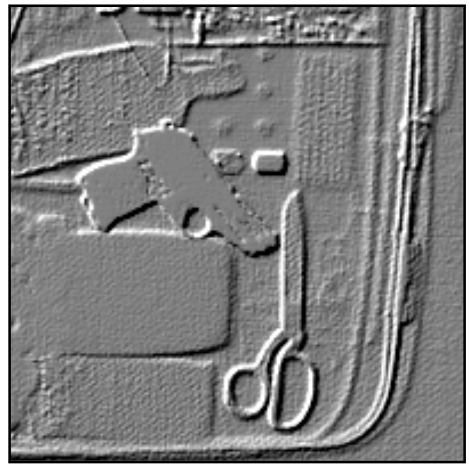
a. Delta function

0	0	0
0	1	0
0	0	0

b. Shift and subtract

0	0	0
0	1	0
0	0	-1





c. Edge detection

-1/8	-1/8	-1/8
-1/8	1	-1/8
-1/8	-1/8	-1/8

d. Edge enhancement

-k/8	-k/8	-k/8
-k/8	k + 1	-k/8
-k/8	-k/8	-k/8

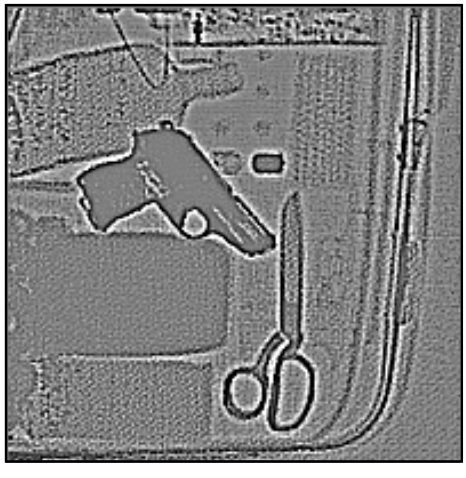




FIGURE 24-4
3×3 edge modification. The original image, (a), was acquired on an airport x-ray baggage scanner. The shift and subtract operation, shown in (b), results in a pseudo three-dimensional effect. The edge detection operator in (c) removes all contrast, leaving only the edge information. The edge enhancement filter, (d), adds various ratios of images (a) and (c), determined by the parameter, *k*. A value of $k = 2$ was used to create this image.

A common image processing technique is shown in (d): **edge enhancement**. This is sometimes called a *sharpening* operation. In (a), the objects have good contrast (an appropriate level of darkness and lightness) but very blurry edges. In (c), the objects have absolutely no contrast, but very sharp edges. The